

Assignment 7: GLMs (Linear Regressios, ANOVA, & t-tests)

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Fall 2025

OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file `<FirstLast>_A07_GLMs.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

!! AI generated code should NOT be used in this assignment

Set up your session

1. Set up your session. Check your working directory. Load the tidyverse, agricolae and other needed packages. Import the *raw* NTL-LTER raw data file for chemistry/physics (NTL-LTER_Lake_ChemistryPhysics_Raw.csv). Set date columns to date objects.
2. Build a ggplot theme and set it as your default theme.

```
#1  
getwd()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --  
## v dplyr      1.1.4      v readr      2.1.5  
## v forcats    1.0.0      v stringr    1.5.1  
## v ggplot2    3.5.1      v tibble     3.2.1  
## v lubridate  1.9.4      v tidyr      1.3.1  
## v purrr      1.0.2
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```

library(agricolae)
library(here)

## here() starts at /home/guest/EDE_Fall2025

library(dplyr)
library(ggthemes)

NTLLitter <- read.csv(here("Data/Raw/NTL-LTER_Lake_ChemistryPhysics_Raw.csv"))

NTLLitter$sampleddate <- as.Date(NTLLitter$sampleddate, format = "%m/%d/%y")

#2
myTheme1 <- theme_base() +
  theme(
    plot.background = element_rect(
      color = 'black',
      fill = 'white'
    ),
    plot.title = element_text(
      color = 'black'
    ),
    axis.text = element_text(
      color = 'black'
    ),
    axis.ticks = element_line(
      color = 'purple',
      linewidth = 1
    )
  )
theme_set(myTheme1)

```

Simple regression

Our first research question is: Does mean lake temperature recorded during July change with depth across all lakes?

3. State the null and alternative hypotheses for this question: > Answer: H0: The mean lake temperature does not change with depth Ha: The mean lake temperature does change with depth
4. Wrangle your NTL-LTER dataset with a pipe function so that the records meet the following criteria:
 - Only dates in July.
 - Only the columns: `lakename`, `year4`, `daynum`, `depth`, `temperature_C`
 - Only complete cases (i.e., remove NAs)
5. Visualize the relationship among the two continuous variables with a scatter plot of temperature by depth. Add a smoothed line showing the linear model, and limit temperature values from 0 to 35 °C. Make this plot look pretty and easy to read.

```

#4
NTL_depthtemp <- NTLLitter %>%
  filter(daynum > 181 & daynum < 213) %>%
  select(lakename, year4, daynum, depth, temperature_C) %>%
  na.omit(NTLLitter)

#5
NTL_plot <- ggplot(NTLLitter, aes(
  x = depth,
  y = temperature_C
))+
ylim(0,35)+
geom_point() +
labs(title = "Temperature by Depth",
  x = "Depth (m)",
  y = "Temperature (C)") +
geom_smooth(method = lm)

print(NTL_plot)

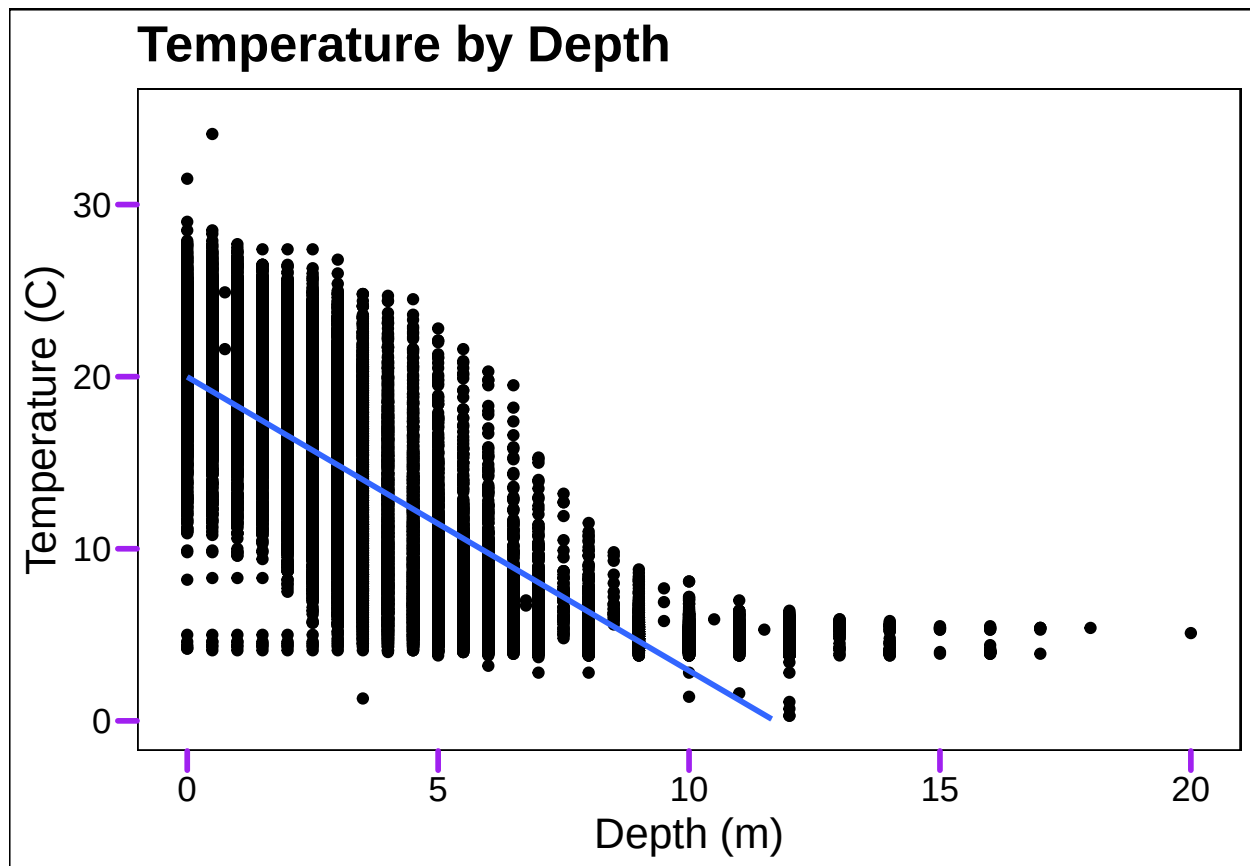
## 'geom_smooth()' using formula = 'y ~ x'

## Warning: Removed 3858 rows containing non-finite outside the scale range
## ('stat_smooth()').

## Warning: Removed 3858 rows containing missing values or values outside the scale range
## ('geom_point()').

## Warning: Removed 33 rows containing missing values or values outside the scale range
## ('geom_smooth()').

```



6. Interpret the figure. What does it suggest with regards to the response of temperature to depth? Do the distribution of points suggest about anything about the linearity of this trend?

Answer: What this scatter plot suggests is that temperature and depth are inversely related, as depth increases, temperature decreases. The distribution of this trend suggests that it is fairly linear, they seem to follow a similar relationship.

7. Perform a linear regression to test the relationship and display the results.

```
#7
NTLregression <- lm(NTL_depthtemp$temperature_C ~ NTL_depthtemp$depth)
cor.test(NTL_depthtemp$temperature_C, NTL_depthtemp$depth)
```

```
##
## Pearson's product-moment correlation
##
## data: NTL_depthtemp$temperature_C and NTL_depthtemp$depth
## t = -165.96, df = 9720, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## -0.8648310 -0.8544571
## sample estimates:
## cor
## -0.8597327
```

```
summary(NTLregression)
```

```
##
## Call:
## lm(formula = NTL_depthtemp$temperature_C ~ NTL_depthtemp$depth)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.5077 -3.0182  0.0743  2.9248 13.6033
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    21.94872    0.06790   323.3  <2e-16 ***
## NTL_depthtemp$depth -1.94700    0.01173  -166.0  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.829 on 9720 degrees of freedom
## Multiple R-squared:  0.7391, Adjusted R-squared:  0.7391
## F-statistic: 2.754e+04 on 1 and 9720 DF,  p-value: < 2.2e-16
```

- Interpret your model results in words. Include how much of the variability in temperature is explained by changes in depth, the degrees of freedom on which this finding is based, and the statistical significance of the result. Also mention how much temperature is predicted to change for every 1m change in depth.

Answer: We can see from the correlation value that depth and temperature are negatively related, -0.86 to be precise, which is a strong negative correlation. The amount of variability in temperature that is explained by depth is .74, the degrees of freedom, which is 9720, and the statistical significance of this result is 2e-16. For every 1 m that depth increases, temperature decreases by -1.947 degrees C.

Multiple regression

Let's tackle a similar question from a different approach. Here, we want to explore what might the best set of predictors for lake temperature in July across the monitoring period at the North Temperate Lakes LTER.

- Run an AIC to determine what set of explanatory variables (year4, daynum, depth) is best suited to predict temperature.
- Run a multiple regression on the recommended set of variables.

```
#9
NTL_AIC <- lm(data = NTL_depthtemp, temperature_C ~ depth + year4 +
              daynum)
step(NTL_AIC)
```

```
## Start: AIC=26016.31
## temperature_C ~ depth + year4 + daynum
##
##           Df Sum of Sq    RSS   AIC
## <none>                 141118 26016
## - year4    1         80 141198 26020
## - daynum   1        1333 142450 26106
## - depth    1       403925 545042 39151

##
## Call:
## lm(formula = temperature_C ~ depth + year4 + daynum, data = NTL_depthtemp)
##
## Coefficients:
## (Intercept)      depth      year4      daynum
##   -6.45556    -1.94726     0.01013     0.04134

#10
NTLmultiregression <- lm(data = NTL_depthtemp, temperature_C ~ depth, year4, daynum)
summary(NTLmultiregression)

##
## Call:
## lm(formula = temperature_C ~ depth, data = NTL_depthtemp, subset = year4,
##     weights = daynum)
##
## Weighted Residuals:
##      Min       1Q   Median       3Q      Max
## -58.629 -32.138  -3.006   35.062   87.158
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  19.42682    0.05686   341.7  <2e-16 ***
## depth       -1.62324    0.00850  -191.0  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 37.51 on 9720 degrees of freedom
## Multiple R-squared:  0.7896, Adjusted R-squared:  0.7895
## F-statistic: 3.647e+04 on 1 and 9720 DF, p-value: < 2.2e-16
```

11. What is the final set of explanatory variables that the AIC method suggests we use to predict temperature in our multiple regression? How much of the observed variance does this model explain? Is this an improvement over the model using only depth as the explanatory variable?

Answer: The final set of explanatory values was all of them, when running my AIC, the results I got said that the AIC value of best fit happened when none of the variables are removed so I kept every variable. The amount of variation explained by this model is .79 which is slightly better than our singular regression model. This is a slight improvement on the previous model because there is more variables to explain variation.

Analysis of Variance

12. Now we want to see whether the different lakes have, on average, different temperatures in the month of July. Run an ANOVA test to complete this analysis. (No need to test assumptions of normality or similar variances.) Create two sets of models: one expressed as an ANOVA models and another expressed as a linear model (as done in our lessons).

#12

```
NTL_ANOVA <- aov(data = NTL_depthtemp, temperature_C ~ lakename)
summary(NTL_ANOVA)
```

```
##              Df Sum Sq Mean Sq F value Pr(>F)
## lakename      8  21214   2651.8   49.04 <2e-16 ***
## Residuals    9713 525188    54.1
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
NTL_lm <- lm(data = NTL_depthtemp, temperature_C ~ lakename)
summary(NTL_lm)
```

```
##
## Call:
## lm(formula = temperature_C ~ lakename, data = NTL_depthtemp)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.766   -6.592   -2.692    7.634   23.832
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      17.6731     0.6741  26.218 < 2e-16 ***
## lakenameCrampton Lake    -2.3212     0.7902   -2.938  0.00332 **
## lakenameEast Long Lake   -7.4054     0.7143  -10.367 < 2e-16 ***
## lakenameHummingbird Lake -6.8998     0.9594   -7.192 6.88e-13 ***
## lakenamePaul Lake        -3.8813     0.6891   -5.633 1.82e-08 ***
## lakenamePeter Lake       -4.3710     0.6878   -6.355 2.18e-10 ***
## lakenameTuesday Lake    -6.6073     0.7002   -9.437 < 2e-16 ***
## lakenameWard Lake        -3.2145     0.9594   -3.350 0.00081 ***
## lakenameWest Long Lake   -6.0876     0.7115   -8.556 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.353 on 9713 degrees of freedom
## Multiple R-squared:  0.03883,    Adjusted R-squared:  0.03803
## F-statistic: 49.04 on 8 and 9713 DF,  p-value: < 2.2e-16
```

13. Is there a significant difference in mean temperature among the lakes? Report your findings.

Answer: There does not appear to be much of a difference between lake temperature during the month of July.

14. Create a graph that depicts temperature by depth, with a separate color for each lake. Add a `geom_smooth` (method = "lm", se = FALSE) for each lake. Make your points 50 % transparent. Adjust your y axis limits to go from 0 to 35 degrees. Clean up your graph to make it pretty.

```
#14.
NTL_plot2 <- ggplot(NTLLitter, aes(
  x = depth,
  y = temperature_C,
  color = lakename
)) +
  ylim(0,35) +
  geom_point(alpha = 0.5) +
  labs(title = "Temperature by Depth",
       x = "Depth (m)",
       y = "Temperature (C)",
       color = "Lake Name") +
  geom_smooth(method = lm, se = FALSE)

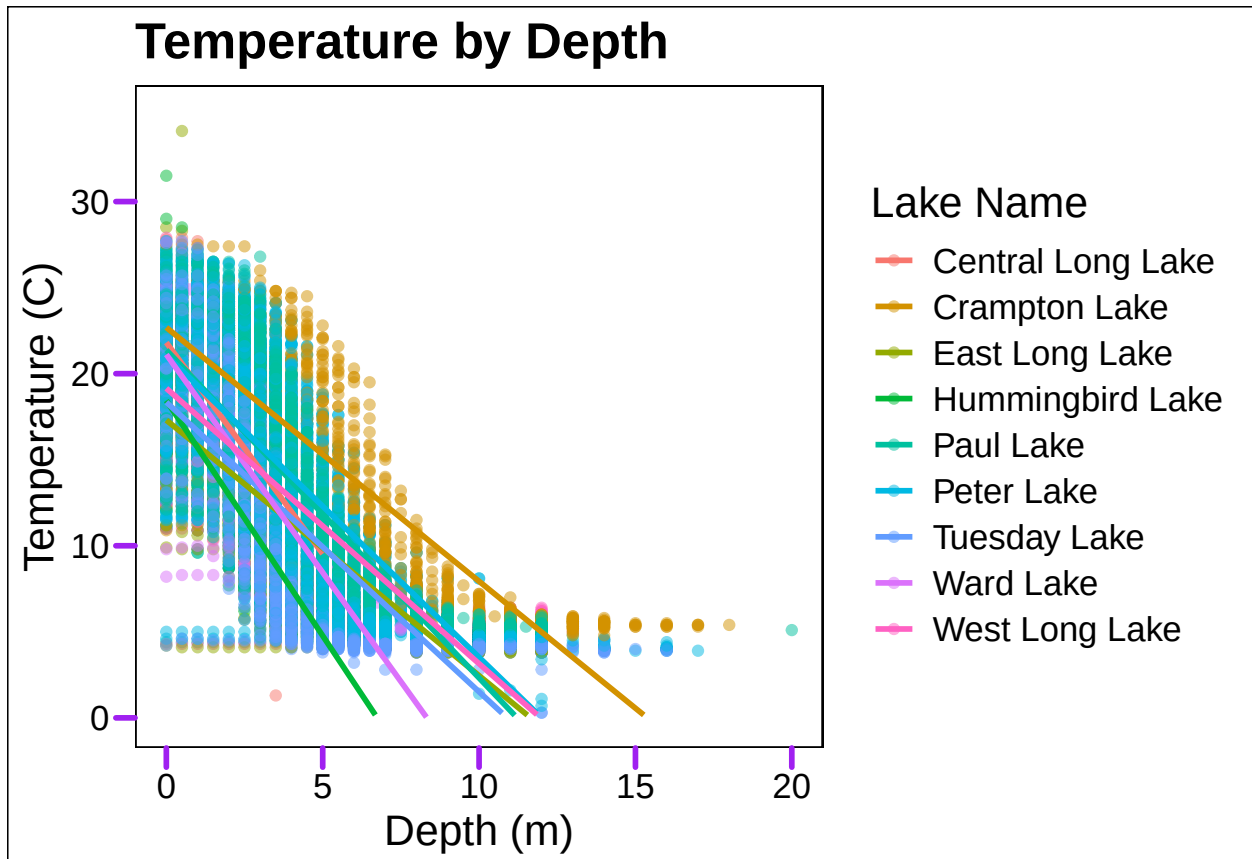
print(NTL_plot2)

## 'geom_smooth()' using formula = 'y ~ x'

## Warning: Removed 3858 rows containing non-finite outside the scale range
## ('stat_smooth()').

## Warning: Removed 3858 rows containing missing values or values outside the scale range
## ('geom_point()').

## Warning: Removed 126 rows containing missing values or values outside the scale range
## ('geom_smooth()').
```



15. Use the Tukey's HSD test to determine which lakes have different means.

#15

TukeyHSD(NTL_ANOVA)

```
## Tukey multiple comparisons of means
## 95% family-wise confidence level
##
## Fit: aov(formula = temperature_C ~ lakename, data = NTL_depthtemp)
##
## $lakename
##
```

	diff	lwr	upr	p adj
## Crampton Lake-Central Long Lake	-2.3212225	-4.7727515	0.1303066	0.0801309
## East Long Lake-Central Long Lake	-7.4054440	-9.6215318	-5.1893561	0.0000000
## Hummingbird Lake-Central Long Lake	-6.8998334	-9.8763946	-3.9232722	0.0000000
## Paul Lake-Central Long Lake	-3.8813120	-6.0191419	-1.7434822	0.0000007
## Peter Lake-Central Long Lake	-4.3710346	-6.5048955	-2.2371736	0.0000000
## Tuesday Lake-Central Long Lake	-6.6072831	-8.7795517	-4.4350145	0.0000000
## Ward Lake-Central Long Lake	-3.2144886	-6.1910498	-0.2379273	0.0229685
## West Long Lake-Central Long Lake	-6.0875867	-8.2949346	-3.8802388	0.0000000
## East Long Lake-Crampton Lake	-5.0842215	-6.5587481	-3.6096949	0.0000000
## Hummingbird Lake-Crampton Lake	-4.5786109	-7.0531008	-2.1041211	0.0000003
## Paul Lake-Crampton Lake	-1.5600896	-2.9141574	-0.2060217	0.0106305
## Peter Lake-Crampton Lake	-2.0498121	-3.3976050	-0.7020192	0.0000841
## Tuesday Lake-Crampton Lake	-4.2860606	-5.6938725	-2.8782488	0.0000000
## Ward Lake-Crampton Lake	-0.8932661	-3.3677559	1.5812237	0.9713958
## West Long Lake-Crampton Lake	-3.7663643	-5.2277226	-2.3050060	0.0000000
## Hummingbird Lake-East Long Lake	0.5056106	-1.7358512	2.7470723	0.9988025
## Paul Lake-East Long Lake	3.5241319	2.6670727	4.3811912	0.0000000
## Peter Lake-East Long Lake	3.0344094	2.1872987	3.8815201	0.0000000
## Tuesday Lake-East Long Lake	0.7981609	-0.1415120	1.7378337	0.1721160
## Ward Lake-East Long Lake	4.1909554	1.9494937	6.4324171	0.0000002
## West Long Lake-East Long Lake	1.3178572	0.2997124	2.3360021	0.0019544
## Paul Lake-Hummingbird Lake	3.0185213	0.8543999	5.1826428	0.0005172
## Peter Lake-Hummingbird Lake	2.5287988	0.3685979	4.6889997	0.0086420
## Tuesday Lake-Hummingbird Lake	0.2925503	-1.9055981	2.4906986	0.9999773
## Ward Lake-Hummingbird Lake	3.6853448	0.6898445	6.6808451	0.0043115
## West Long Lake-Hummingbird Lake	0.8122467	-1.4205745	3.0450678	0.9700210
## Peter Lake-Paul Lake	-0.4897225	-1.1036180	0.1241730	0.2442990
## Tuesday Lake-Paul Lake	-2.7259711	-3.4623514	-1.9895907	0.0000000
## Ward Lake-Paul Lake	0.6668235	-1.4972980	2.8309450	0.9895659
## West Long Lake-Paul Lake	-2.2062747	-3.0404749	-1.3720745	0.0000000
## Tuesday Lake-Peter Lake	-2.2362485	-2.9610258	-1.5114713	0.0000000
## Ward Lake-Peter Lake	1.1565460	-1.0036549	3.3167469	0.7703831
## West Long Lake-Peter Lake	-1.7165522	-2.5405279	-0.8925764	0.0000000
## Ward Lake-Tuesday Lake	3.3927945	1.1946462	5.5909429	0.0000597
## West Long Lake-Tuesday Lake	0.5196964	-0.3991749	1.4385677	0.7121762
## West Long Lake-Ward Lake	-2.8730982	-5.1059193	-0.6402770	0.0021521

16. From the findings above, which lakes have the same mean temperature, statistically speaking, as Peter Lake? Does any lake have a mean temperature that is statistically distinct from all the other lakes?

Answer: Ward Lake, Paul Lake, and Crampton Lake have fairly similar averages to Peter Lake. The one lake that has distinct mean temperature is Central Long Lake.

17. If we were just looking at Peter Lake and Paul Lake. What's another test we might explore to see whether they have distinct mean temperatures?

Answer: Another test we could run would be a t-test.

18. Wrangle the July data to include only records for Crampton Lake and Ward Lake. Run the two-sample T-test on these data to determine whether their July temperature are same or different. What does the test say? Are the mean temperatures for the lakes equal? Does that match you answer for part 16?

```
July_lakes <- NTLLitter %>%
  filter(lakename == c("Crampton Lake", "Ward Lake"))

tTest <- t.test(July_lakes$temperature_C ~ July_lakes$lakename)
tTest

##
## Welch Two Sample t-test
##
## data: July_lakes$temperature_C by July_lakes$lakename
## t = 3.4458, df = 546.02, p-value = 0.0006131
## alternative hypothesis: true difference in means between group Crampton Lake and group Ward Lake is not equal to 0
## 95 percent confidence interval:
##  0.7354581 2.6857018
## sample estimates:
## mean in group Crampton Lake      mean in group Ward Lake
##           14.16486                12.45428
```

Answer: These tests show the mean temperatures in Crampton and Ward lake, their means are slightly different, off by about a degree and a half. This matches our answer in 16 because it shows that both values are close and so when compared to Peter Lake, they would have very similar differences to Peter's mean.